Atomarine

Cheap Nuclear Propulsion as a Service without Regulatory Roadblocks

Concept introduction and motivation

Nuclear propulsion has been technically proven for decades. It eliminates carbon-cost exposure, enables higher sailing speeds that lift freight revenue per ship, and avoids refuelling for years, resulting in fuel-cost stability and operational time savings.

However, installing a reactor on a cargo ship faces several roadblocks:

- Port access, public opposition, and insurance barriers
- Challenging economics driven by high reactor cost
- Near-decade-long reactor licensing timelines compounded by multiple designs and jurisdictions
- A steep adoption burden for shipowners: specialized staff, large nuclear CAPEX, and continuous compliance with strict nuclear regulation

To address these issues, Atomarine is developing a detachable nuclear power-sharing model in which standardized nuclear vessels rendezvous with conventional merchant ships in international waters and supply up to 99% of propulsion power for the voyage via ship-to-ship offshore dynamic cables.

Economic benefits:

- Lower propulsion cost through economies of scale
- High utilization of the nuclear asset
- High speeds without carbon exposure
- Fuel cost stability through multi-year refueling intervals

Regulatory benefits:

- Improved safety case: nuclear vessels are kept away from population centers and the design is optimized for safety/security without cargo tradeoffs.
- Avoid port access and multi-national regulatory roadblocks: nuclear vessels remain in international waters.

Easy adoption into today's industry:

- Propulsion as a service: no nuclear CAPEX for shipowners; no nuclear compliance burden; no crew retraining.
- Standardized, mature nuclear tech: PWR-based design licensed once; scalable via existing supply chains.
- Fleet-compatible: ships join via retrofit to hybrid electric; existing fleet is valorized; no long-term fuel lock-in.

Brief overview of the technology

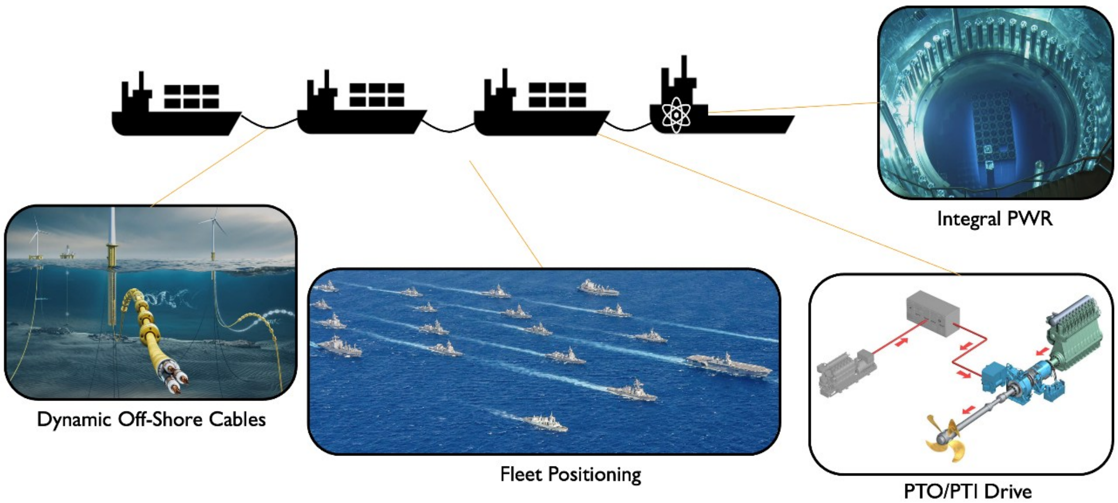

Figure 1 Schematic overview of the main technology components. Note that a convoy will have a hub-and-spoke cable architecture, rather than the pictorial serial configuration.

Reactor: Gen III+ PWR (iPWR preferred)

- Gen III+ PWR to leverage regulator familiarity + existing supply chains
- Operating base: 160+ naval PWR vessels, ~2,000 reactor-years, plus Russian icebreakers
- iPWR preferred (not required): smaller/lighter; no LOCA risk
- Multiple PWR vendors (US/UK/France/S. Korea) and existing starting designs
- Vendor depth + existing starting designs is a major advantage

Ship integration: hybrid PTI/PTO

- Indirect electric drive ships: plug-in → no change to propeller, motor, drivetrain
- Direct-drive ships: motor-generator insertion (PTI) on shaftline → hybrid propulsion
- PTI retrofits are common; indicative cost ~$2–3M

Power transfer: offshore dynamic cables

- Dynamic offshore HV cables and handling mature from floating offshore wind
- Relative motion handled by proven offshore systems: active heave compensation + load balancing
- Cable-laying experts indicate workable; cable motion adds cooling; two flexible endpoints help vs fixed ground tie

Fleet positioning

- Several 100m spacing between ships to reduce collision risk
- Autonomy can simplify scaling (hub-and-spoke vision); not required, but helpful

Summary of the economic modeling

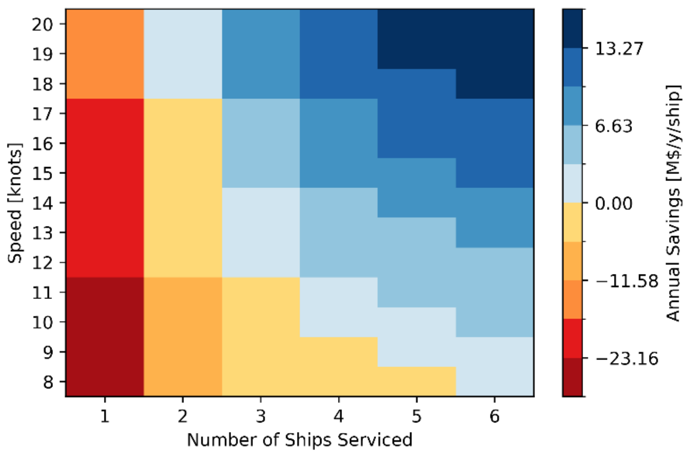

Figure 2 Illustrative annual savings per vessel when using Atomarine power under various operating modes and 2040 carbon taxes.

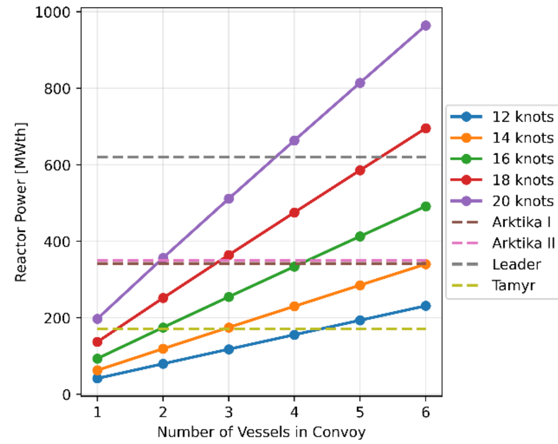

Figure 3 Thermal reactor power as a function of speed and convoy size compared to the thermal power of the Russian nuclear icebreakers

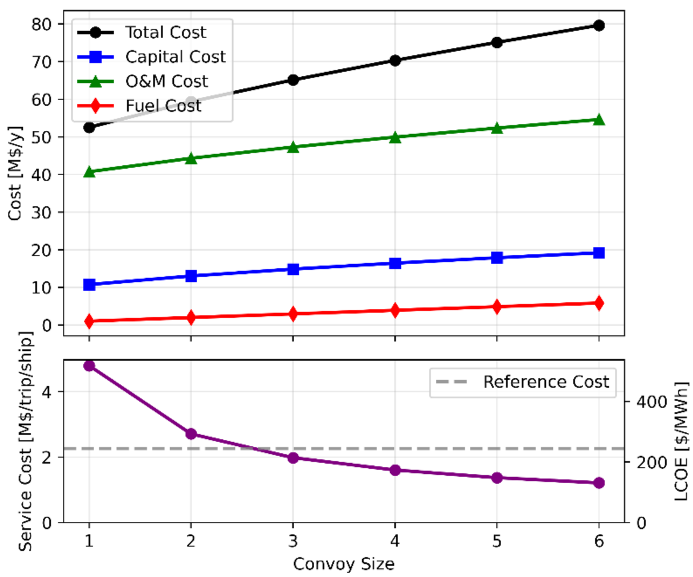

Figure 4 Illustrative breakdown of the total annualized nuclear vessel cost along its main components as a function of the convoy size for a cruising speed of 12 knots

Main assumptions of the shown scenario:

- Ref. bulker speed: 10.5 knots
- Trip length: 4600 NM
- Reference TCE: 27500 $/d
- Future fuel cost: 1000 $/mt
- Deployment year: 2040
- Fleet size: 50
- Nuclear vessel cost: 0.3-2.8 B$
  (dependent on required power)
- Onboard crew: 54
- Core lifetime: 5 years
- Fuel enrichment: 5%
- WTP for FuelEU surplus: 33%
- EUA price: 108 $/ton

Our economic modeling uses a Capesize bulker on the Bolivar–Rotterdam route, aligned with the Baltic Exchange Capesize reference routes (C8–C9). Ongoing modeling focuses on container ships due to their fixed schedules and higher operational speeds.

The economics of the power-sharing concept are strongly driven by convoy size and operating speed, with high-speed, multi-ship operation favored (Figure 2). Annual savings per vessel can reach single-digit millions under low-speed operating modes, while doubledigit millions per year generally require either high speed or larger convoys.

Increasing either cruising speed or the number of ships per convoy raises the required reactor power. In all cases, the required power remains comparable to Russian nuclear icebreakers (Figure 3). Because the Atomarine ship carries no cargo, it can be small and requires only about a quarter of the power of a single Capesize bulker.

As reactor power increases, cost per unit energy decreases due to economies of scale. As shown in Figure 4, the main cost driver for maritime nuclear is staffing cost, primarily driven by onboard staffing requirements. Crucially, the required onboard staff does not increase with reactor power: doubling reactor power does not require more security staff, operators, engineers, or similar roles. There are also economies of scale in capital cost due to sublinear scaling. Overall, the LCOE remains high relative to onshore nuclear, as expected for maritime nuclear, given the small reactor size and fixed staffing. However, this is addressed by serving a convoy simultaneously.

Higher convoy speeds increase profitability by increasing freight revenue per ship and generating more FuelEU surplus credits that can be sold through pooling. Overall, results are only moderately sensitive to the deployment year, because FuelEU surplus credit sales smooth the impact over time.